Research Article | Vol. 24, No. 22 | 31 Oct 2016 | OPTICS EXPRESS A1300

Optics EXPRESS

Self-reverse-biased solar panel optical receiver for simultaneous visible light communication and energy harvesting

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Abstract: We propose a self-reverse-biased solar panel optical receiver for energy harvesting and visible light communication. Since the solar panel converts an optical component into an electrical component, it provides both energy harvesting and communication. The signal component can be separated from the direct current component, and these components are used for communication and energy harvesting. We employed a self-reverse-biased receiver circuit to improve the communication and energy harvesting performance. The reverse bias on the solar panel improves the responsivity and response time. The proposed system achieved 17.05 mbps discrete multitone transmission with a bit error rate of 1.1 x 10^{-3} and enhanced solar energy conversion efficiency.

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OCIS codes: (060.4510) Optical communications; (040.5350) Photovoltaic.

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1. Introduction

In recent years, data traffic problem has come to the fore owing to increasing data traffic requirements [1]. Visible light communication (VLC) is researched as a method to solve this problem. VLC can complement wireless communication to alleviate data traffic problem

#268975 http://dx.doi.org/10.1364/OE.24.0A1300
Journal © 2016 Received 22 Jun 2016; revised 19 Aug 2016; accepted 26 Aug 2016; published 7 Sep 2016

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because it uses a wide bandwidth of the visible light spectrum and it does not interfere with other types of wireless communication. Therefore, VLC can supplement and coexist with wireless communication [2,3].

Light-emitting diodes (LEDs) are the foremost candidates for next-generation illumination because of their long lifetime and low power consumption compared with incandescent or fluorescent lamps. For these reasons, they are used for simultaneous illumination and communication as a transmitter in VLC systems [4].

Despite these advantages, VLC has a low data rate because of the bandwidth limitation of LEDs. To increase the bandwidth of LEDs, blue filtering and pre- and post-equalizing were adopted [5]. In addition, orthogonal frequency division multiplexing (OFDM) was used to improve the spectral efficiency [6,7]. Higher data rates can be achieved using wavelength division multiplexing and multi-input multi-output systems [8,9].

In VLC systems, photodiodes (PDs) are typically used as a receiver because of their fast response time. However, PDs require external power for operation and a convex lens for gathering light. To overcome these inconveniences, a solar panel can be used as a receiver. Both the solar panel and PD convert light energy to electric energy. The biggest difference between the two is the receiving area. Consequently, a solar panel does not need a convex lens because its receiving area is large enough to receive sufficient light without one. In addition, a solar panel also does not need external power for operation because its built-in voltage is sufficient to separate electron–hole pairs in the depletion region.

However, a solar panel is not appropriate for use as an optical receiver because of its slow response time. The capacitance of an element is one of the factors affecting its response time, and solar panels have a large capacitance owing to their large receiving area [10]. Despite this disadvantage, if a solar panel can harvest energy and perform communication simultaneously, it becomes attractive for use as an optical receiver. The use of a solar panel as an optical receiver for optical wireless communications with simultaneous energy harvesting was reported recently [11–13]. A special receiver circuit was employed for communication with simultaneous energy harvesting [13]. However, the slow response time and the low harvested energy of solar panels are still challenges for communication and energy harvesting. Therefore, it is necessary to improve both communication performance and solar energy conversion efficiency for various applications. To solve these challenges, we propose a novel technique which enhances communication and energy harvesting efficiency simultaneously.

In this paper, we propose a solar panel design based on a self-reverse-biased optical receiver for simultaneous VLC and energy harvesting. We theoretically and experimentally demonstrate how a reverse bias improves the performance of the solar panel. The factors that affect the response time and responsivity of the solar panel are explained. In addition, the frequency response and responsivity of the solar panel are measured for various reverse biases. A receiver circuit with a self-reverse bias is also proposed for simultaneous communication and energy harvesting. The parameters of each element that maximize performance are determined through simulation. The transmission performance of the proposed system is experimentally demonstrated up to 17 Mbps. Using the proposed receiver circuit improves both the energy harvesting and communication performance.

2. Effects of reverse bias on solar panel

Three variables determine the response time of a PD: the drift time of photocarriers inside the depletion region, the diffusion time of photocarriers outside the depletion region, and the capacitance of the PD. A PD usually requires a reverse bias for operation. When a higher reverse bias is applied, fewer photocarriers are generated outside of the depletion region. In other words, most of the photocarriers are generated inside the depletion region. Because a higher reverse bias is applied, the depletion region becomes wider. Therefore, the responsivity increases slightly because more photocarriers are generated in the depletion

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region. In addition, because the electric field is stronger in the depletion region, the carrier drift velocity increases. The drift time t_d is given by

$$t_d = \frac{w}{v_d}. \tag{1}$$

where w is the width of the depletion region, and v_d is the drift velocity. A high carrier drift velocity results in a short drift time; consequently, the response time becomes shorter. Because the structure of solar panels is similar to that of PDs, when a reverse bias is applied to a solar panel, its response time becomes shorter, and its responsivity increases.

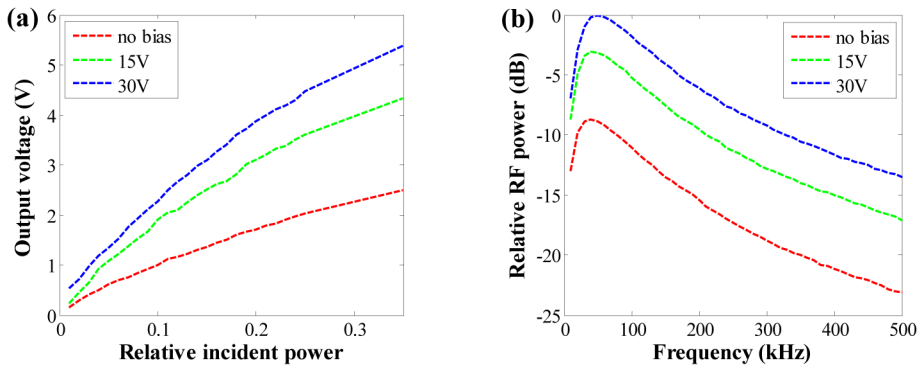

Fig. 1. Output voltage of solar panel versus (a) incident light power and (b) frequency response of solar panel for a reverse bias of 0, 15, and 30 V.

Figure 1(a) shows the measured output voltage of a solar panel versus the incident light power for a reverse bias of 0, 15, and 30 V. We see that the responsivity increases with increasing reverse bias. Figure 1(b) shows the measured frequency response of a solar panel for a reverse bias of 0, 15, and 30 V. Scheme of measuring frequency response consist of the LED and the solar panel. The frequency response increases with increasing reverse bias owing to the increased responsivity. In addition, the 3-dB bandwidth increases from 90 to 120 kHz when we increase the reverse bias from 0 to 30 V. Therefore, we can expect to enhance both the energy harvesting and communication performance by applying a reverse bias to the solar panel.

3. Self-reverse-biased solar panel receiver circuit

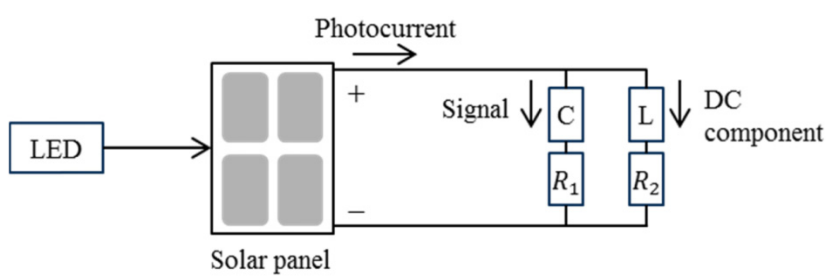

Fig. 2. Solar panel receiver circuit for simultaneous energy harvesting and communication.

To generate signal light in an LED, a DC bias is conventionally combined with the signal component using a bias tee. On the other hand, we can separate the signal and DC components of received light using a bias tee in the reverse process. Figure 2 shows the proposed receiver circuit for using a solar panel as a receiver for communication with simultaneous energy harvesting [13]. The received signal and DC components are separated

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through a capacitor and an inductor, where the inductor attenuates the signal, and the capacitor removes the DC component. Therefore, they can be extracted from each load resistor. The voltage at the load resistor R_1 is expressed as

$$V_{R_1} = \frac{R_1}{R_1 + 1/jwC} ZI. \tag{2}$$

In Eq. (2), I is the generated photocurrent at the solar panel, and Z is the impedance of the receiver circuit. Consequently, ZI is the voltage of the entire circuit. Z is expressed as

$$Z = (\frac{1}{jwC} + R_1) // (jwL + R_2). \tag{3}$$

Therefore, the frequency response of the receiver circuit is given by

$$|H(jw)|_{receiver.circuit} = \sqrt{\frac{(R_1 R_2)^2 + (wR_1 L)^2}{(R_1 + R_2)^2 + (wL - \frac{1}{wC})^2}}. \tag{4}$$

The overall frequency response of the system is expressed as

$$|H(jw)|_{overall.system} = |H(jw)|_{solar.panel} |H(jw)|_{receiver.circuit}. \tag{5}$$

By using Eq. (5), the optimal values of the parameters R_1, R_2, C and L for communication can be simulated. We found that $R_1 = 10000\,\Omega$, $R_2 = 10\,\Omega$, $C = 1\,\text{uF}$, and $L = 40\,\text{mH}$. These optimization can be changed depending on the type of solar panel.

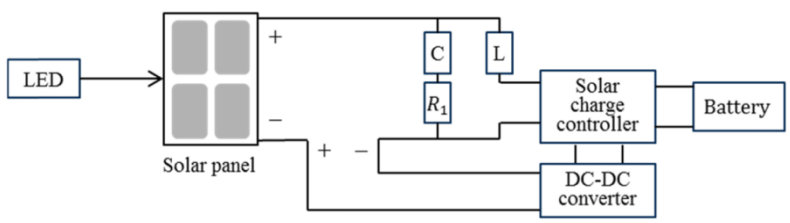

Fig. 3. Solar panel based on self-reverse-biased receiver circuit for simultaneous energy harvesting and communication.

Applying a reverse bias to the solar panel improves the performance, as explained in the previous section. We propose an additional receiver circuit for applying a self-reverse bias without external power by using energy harvested by the solar panel (Fig. 3). First, the energy harvesting branch is connected to a solar charge controller to supply the rated current and rated voltage to the battery. Next, the solar charge controller is connected to the battery for charging. Since high voltage is needed for the reverse bias, a DC–DC converter is necessary. After DC–DC conversion, a high DC voltage is generated and applied as the reverse bias.

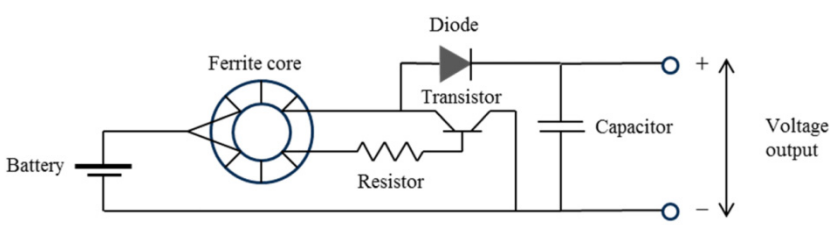

Fig. 4. Joule thief DC–DC up-converter.

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A joule thief is used as the DC–DC up-converter [14]. Figure 4 shows the joule thief circuit. It consists of a ferrite core, a 1 kΩ resistor, and a transistor and converts a low voltage to a high voltage without external power. Because the output voltage has large ripples, a capacitor is used to remove the ripples. The conversion rate is determined according to the number of turns. We obtained a 30 V output voltage from a 2 V battery voltage.

4. Experiment and results

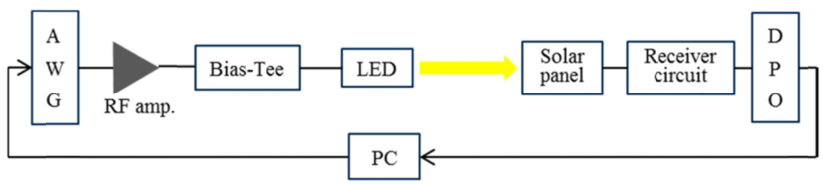

Fig. 5. Experimental setup.

Figure 5 shows the experimental setup for verifying the communication performance of the proposed system based on a solar panel in a VLC system. An arbitrary signal is generated by MATLAB, and an arbitrary waveform generator (AWG) generates this signal electrically. The output of the AWG is amplified by an electrical amplifier. A commercial white LED transfers the amplified signal, which is contained within the light. The transmission distance is 10 cm. A solar panel (POW112D2P) is used as an optical receiver. It receives this light, and the receiver circuit separates the light into signal and DC components. A reverse bias is supplied using the energy harvested from the DC component. A digital phosphor oscilloscope (DPO) was used for conversion of the electrical signal to a digitized signal and for demodulation.

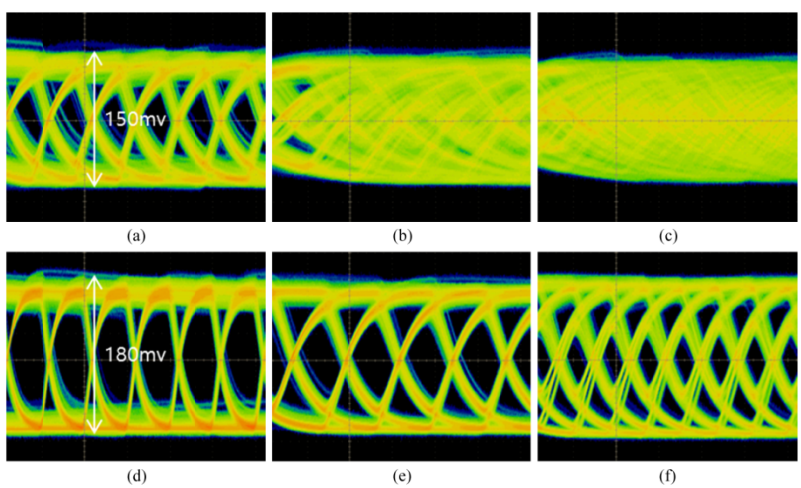

Fig. 6. Eye diagrams with (a) no bias and 1 Mbps OOK, (b) no bias and 2 Mbps OOK, (c) no bias and 3 Mbps OOK (d) 30 V reverse bias and 1 Mbps OOK, (e) 30 V reverse bias and 2 Mbps OOK and (f) 30 V reverse bias and 3 Mbps OOK.

Firstly, to verify the enhancement of the response time and signal power, eye diagrams were measured, as shown in Fig. 6. To measure the eye diagrams, a simple non-return-to-zero on-off-keying modulation format was used for the data. Figures 6(a)-6(c) and Figs. 6(d)–6(f) show the eye diagrams with no reverse bias and a 30 V reverse bias, respectively. The reverse bias helps decrease the rising and falling times. In addition, the signal power is increased from 150 to 180 mV. Therefore, we obtained better eye opening when the reverse bias was applied.

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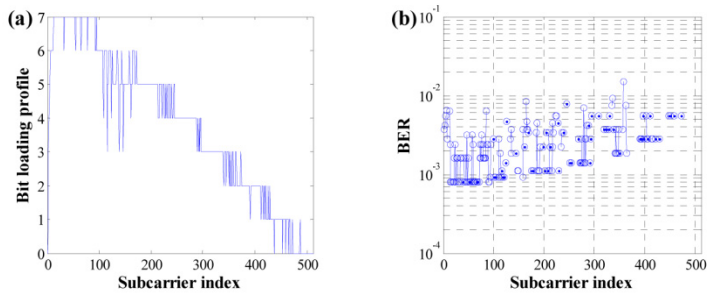

Fig. 7. Transmission performance of DMT modulation: (a) bit loading profile, (b) BER.

Next, to estimate the VLC transmission performance using the proposed solar panel receiver circuit, transmission measurements were made using discrete multitone (DMT) modulation. Serial arbitrary streams are generated in MATLAB and converted into parallel streams, which are mapped into 4-quadrature amplitude modulation. Before the parallel streams were converted into serial streams, a cyclic prefix (CP) was inserted. At the receiver, the received signal is converted into parallel streams and the CP is extracted. The DMT signal is regenerated following bit loading based on the signal-to-noise ratio of each subcarrier. We use 512 subcarriers for the DMT signal. The signal bandwidth is 9 MHz, which is 75 times larger than the 3-dB bandwidth of the solar panel. The bit error rate (BER) target of bit loading is 2×10^{-3}. Figure 7 shows the overall communication performance of the system. The bit loading profile is formed according to the overall frequency response of the system, as shown in Fig. 7(a). Almost all of the subcarriers have BERs below 2×10^{-3}, as shown in Fig. 7(b). A maximum data rate of 17.05 Mbps with a BER of 1.1×10^{-3} was achieved.

To estimate the effect of a reverse bias on energy harvesting, the solar energy conversion efficiency was measured. A solar energy conversion efficiency of 8% was measured with 250 lx when no bias was applied. However, the solar energy conversion efficiency increased up to 26% with 250 lx when a 30 V reverse bias was applied. It means even if a portion of harvested energy is consumed for a reverse biasing, overall energy harvesting efficiency increases compare with the case of no reverse bias. The reason is that most photocarriers are generated inside the depletion region, and the strengthened electric field causes the carriers to be collected across the junction well when the reverse bias is applied.

Since irradiance is in inverse proportion to square of the distance between the solar panel and the LED, if the distance increases, system performance will decrease. However energy harvesting from ambient light such as solar light is constant. In addition, saturation point of the solar panel increases when reverse bias is applied. Therefore, the proposed system would be robust in outdoor environment. These results confirm that our proposed system also provides a feasibility of enhanced solar energy technology.

5. Conclusion

In summary, we have proposed a self-reverse-biased solar panel receiver using harvested energy for energy harvesting and VLC. Because the modulated light can be separated into signal and DC components by using a solar panel, simultaneous energy harvesting and communication were possible. Applying a reverse bias to a solar panel using the proposed receiver circuit improves the responsivity and response time. In an experimental demonstration, we achieved 17.05 Mbps DMT transmission with a BER of 1.1×10^{-3}. In addition, the solar energy conversion efficiency increased from 8% to 26% with 250 lx. These results imply that a solar panel-based optical receiver is feasible for use as an optical receiver in a self-powered smart device for potential applications in the Internet of Everything.